Corrected MCQs

1. **C.** Dirty $x^3/3$ give back x^2. Derivating x^2 is not looking for a antiderivative.
2. **B.** The factor 1/2 compensates for factor 2 of the compound bypass. C is a real constant.
3. **D.** It's the form u'/u, $u = 1 + x^2 > 0$ everywhere. x is not defined on all $\mathbb{R}$.
4. **A.** The difference between antiderivatives is constant; this translates into a vertical translation.
5. **C.** The antiderivatives are $x^2 + C$. The condition gives $1 + C = 3$, so $C = 2$.
6. **B.** The solutions are Ce^{-2t} ; the initial condition gives $C = 3$. An affine function is not suitable Every moment.

7. **D.** A constant has a null derivative; solve 0 $= -2L + 6$ given $L = 3$.
8. **A.** Subtracting the two equations eliminates the second member. This is the associated homogeneous equation.
9. **C.** The update is $y_1 = y_0 + h(-y_0) = 2 - 2h$. Euler gives an approximation, distinct from the exact solution 2 e^{-h}.
10. **B.** The derivative is in degrees Celsius per minute: k multiplies a temperature difference and a For unit min $^{-1}$.

True/false. False: adding a constant gives another antiderivative. False: for $y' = -y, y(0) = 2,$ Step 1 gives 0, instead of 2/ e. True, on the required interval and with the required differentiability.

Corrected applications and support

Application 1. $x^4 - x^2 + x$; $-e^{-3x}/3$; $\frac{1}{2}\ln|2x+1|$ on each of the intervals not containing $-1/2$. Derivatives give back the announced expressions.

Application 2. $F = x^2 + C$ and $F(1) = 3$ give C=2. On the two half-rights, $\ln|x| + C_-$ for $x < 0$ and $\ln|x| + C_+$ for $x > 0$ is a antiderivative, even if $C_- \neq C_+$.

Implementation 3. $y = 5e^{-2t}$. The null function satisfies the equation alone, but not the condition $y(0) = 5$.

Implementation 4. Constant 2 is a solution. $z = y - 2$ check $z' = 2z$ and $z(0) = 1$, therefore $y = 2 + e^{2t}$. Its derivative $2e^{2t}$ is good 2 $y - 4$ and $y(0) = 3$.

Application 5. Check e^t shows only that it is a solution. Functions 2 e^t, $-e^t$ and the for completeness, starting from any solution and establishing that it a the form announced.

Implementation 6. Ongoing: $e^{-0{,}1t} = 1/2$, or $t = 10\ln 2 \simeq 6{,}93$ min. Euler: $T_n = 20 + 60(0{,}8)^n$; $(0{,}8)^3 = 0{,}512 > 1/2$, $(0{,}8)^4 = 0{,}4096 < 1/2$. The first point of the grid is therefore 8 min. imposed discrete moments and Euler introduced an error of approximation.

Research and comparison

Calculate a first step by hand, then compare h=2 and h=1 with the program provided.
The observed convergence of an approximation does not replace an estimate of its error.
Do not call "exact solution" a list of Euler values.

Conduct and correction

Benchmarks for the teacher

Distinguish unknown function and actual variable. Have each antiderivative checked by bypass. For $y' = ay + b$, first seek the constant solution. The case $a = 0$ must remain visible. Trigonometric antiderivatives are given here as reference table, but their use Evaluated waits for the trigonometry sequence. To Euler, keep the units and compare to a known exact solution.

Corrected activity

1. The equation is $T' = -0{,}1T + 2$. A constant solution is 20; all solutions are $T(t) = 20 + Ce^{-0{,}1t}$. Like $T(0) = 80$, $C = 60$. Verification $T'(t) = -6e^{-0{,}1t} = -0{,}1(T(t) - 20)$ and $T(0) = 80$.

2. $T'(t) = -6e^{-0{,}1t} < 0$: temperature decreases strictly. $e^{-0{,}1t} \to 0$ in $+\infty$, $T(t) \to 20\,°\text{C}$, remaining above 20 °C.

3. $T(t) = 40 \iff e^{-0{,}1t} = \frac{1}{3}$. Taking the logarithm, $-0{,}1t = -\ln 3$, either $t = 10\ln 3 \simeq 10{,}9861$ min. Strict decline ensures that it is the only moment of transition to 40 °C.

4. For $h = 1$, $Y_{k+1} = Y_k - 0{,}1(Y_k - 20)$. Thus $Y_1 = 80 - 6 = 74$ and $Y_2 = 74 - 5{,}4 = 68{,}6$. The solution exact given $T(2) = 20 + 60e^{-0{,}2} \simeq 69{,}1238$. To underestimate here the temperature of about $0{,}5238\,°\text{C}$.

5. The model assumes an ambient temperature of 20 °C and a constant exchange coefficient of $0{,}1\ \text{min}^{-1}$. A change of part, lid or agitation may alter these assumptions; the formula is no longer automatically adapted.

E1. Linearity gives $F(x) = x^3 - 2x^2 + x + C$, $C \in \mathbb{R}$. We're checking. $F'(x) = 3x^2 - 4x + 1$. All antiderivatives on the interval $\mathbb{R}$ different from a constant.

E2. The denominator is positive and its derivative is $2x$. The antiderivatives are $\ln(x^2+4) + C$. The condition in 0 gives $C = -$ No 4, therefore. $F(x) = \ln(x^2+4) - 4$; check $F(0) = 0$ and $F'(x) = \frac{2x}{x^2+4}$.

E3. The solutions of $y' = 3y$ are Ce^{3x}. The condition $Ce^3 = 2$ requires $C = 2e^{-3}$. Thus $y(x) = 2e^{3(x-1)}$, of which the derivative is $3y(x)$ and the value in 1 is 2.

E4. Constant 4 is a particular solution, so the solutions are 4 $+ Ce^{-x}$. Like 4 $+ C = 7$, $C = 3$. The solution is $y(x) = 4 + 3e^{-x}$; its derivative $-3e^{-x}$ is good $-y + 4$, and $y(0) = 7$.

E5. $g(x) = x + 1$ check $g' = 1 = g - x$. If $z = y - g$, then $z' = z$, therefore $y = x + 1 + Ce^x$. The condition $y(0) = 0$ tax $C = -1$. One obtains $y = x + 1 - e^x$ and we're checking $y' = 1 - e^x = y - x$.

Corrected from DM7

1. If $g(t) = at + b$, then $g' = a$ and $-g + 2t + 1 = (2 - a)t + 1 - b$. For identity to be true to everything t, it takes $2 - a = 0$ and $a = 1 - b$. So $a = 2$, $b = -1$, and $g(t) = 2t - 1$.

2. Poisons $z = y - g$. By subtracting the two equations, $z' = -z$, therefore $z = Ce^{-t}$. Thus $y(t) = 2t - 1 + Ce^{-t}$. Like $y(0) = 3$, $C = 4$. The function $y(t) = 2t - 1 + 4e^{-t}$ Check it out. $y' = 2 - 4e^{-t} = -y + 2t + 1$ and $y(0) = 3$.

3. $y'(t) = 2 - 4e^{-t}$. This derivative is negative for $t < \ln 2$, none in $\ln 2$, positive after. The solution decreases on $[0, \ln 2]$ then grows on $[\ln 2, +\infty[$. Its minimum is $y(\ln 2) = 2\ln 2 - 1 + 2 = 1 + 2\ln 2$.

4.
```
def euler_dm7(N):
    h = 2/N
    y = 3.0
    for k in rows (N):
        t = k*h
        y = y+h*(-y+2*t+1)
    return y
```

The time used in the second member is that of the beginning of the step.
$Y_{10} \simeq 3{,}42950$ and $Y_{100} \simeq 3{,}53048$, while $y(2) = 3 + 4e^{-2} \simeq 3{,}54134$. Absolute errors are valid about 0 ,11184 and 0,01086. To verify these values without loop, posisons $Z_k = Y_k - (2kh - 1)$. The recurrence of Euler gives $Z_{k+1} = (1 - h)Z_k$ and $Z_0 = 4$. Therefore $Y_N = 3 + 4(1 - \frac{2}{N})^N$.

5. A differential equation solution shall check for equality of functions throughout the interval, and the initial condition. Some numerical values control only a finite number of
The symbolic substitution of question 2 is a verification of the whole field.

Corrected work in the manual or an anal

Exercise 30. If y is constant equal to ℓ, the equation becomes $0 = -\frac{1}{2}\ell + 3$, so $\ell = 6$. By posing $z = y - 6$, we get $z' = -\frac{1}{2}z$; all solutions are $y(x) = 6 + Ce^{-\frac{x}{2}}$, $C \in \mathbb{R}$.

Exercise 25.

1. $f' = \frac{1}{2}f$ gives $f(x) = Ce^{\frac{x}{2}}$. Like $f(0) = 1$, $C = 1$.
2. $5f' = f$ gives $f(x) = Ce^{\frac{x}{5}}$. The condition $f(1) = 0$ tax $Ce^{\frac{1}{5}} = 0$; the exponential being never None, $C = 0$. The solution is the null function.
3. $f' + f = 0$ given $f(x) = Ce^{-x}$, therefore $f'(x) = -Ce^{-x}$. The condition relates to the derivative: $f'(1) = 2$ is equivalent to $-Ce^{-1} = 2$, or $C = -2e$. Thus $f(x) = -2e^{1-x}$.